

Exploring uncertainty in Antarctica's future sea level contribution

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Chapter 1

Literature Review

1.1 Sea level

1.1.1 Timescales of sea level change

NOTE: Long version of this section, which is less information dense and more readable, but also maybe a bit waffley.

Sea level is not a single measurement, but is rather a dynamic variable that changes over a range of temporal and spatial scales (Woodworth et al., 2019). It is helpful to distinguish between 'local' and 'global' sea level change, and to carefully define the timescales over which we are considering sea level variability (Gregory et al., 2019). For the purposes of this thesis, we will be primarily concerned with global mean sea level change over timescales greater than a decade, but we will briefly review the processes that drive sea level variability on all timescales, as these are important for understanding the impact on coastal communities.

Sea level change can also be categorised by the types of process that drive it. Since ocean volume is determined by its density and its mass, sea level

change can be expressed as a combination of changes in these two quantities. 'Steric' sea level is associated with ocean density, whilst 'barystatic' sea level is associated with its mass (Gregory et al., 2019). Steric sea level can be further subdivided into 'thermosteric' effects, driven by ocean temperature, and 'halosteric' effects, driven by ocean salinity. Barystatic sea level change obeys conservation of mass in the Earth system, and can therefore be calculated as the mass flux imbalance between the ocean and the ice sheets, glaciers, and land water storage.

On short timescales (minutes to days), sea level is dominated by the astronomical tide. However, other processes such as storm surges, coastal waves, and meteotsunamis can contribute significantly to sea level variability (Pugh and Woodworth, 2014). On intermediate timescales (weeks to years), steric effects associated with the seasonal cycle become relevant (Gill and Niller, 1973). Changes in ocean circulation and interannual climate variability also drive sea level variability on these timescales (Bingham and Hughes, 2009; Han et al., 2017).

On longer timescales (decades to millennia), sea level is influenced by changes in the Earth's climate and in the solid Earth (Fox-Kemper et al., 2021). Climate change (natural or anthropogenic) drives changes in sea level through a combination of thermosteric effects (ocean warming), barystatic effects (melting of ice sheets and glaciers), and the consequent halosteric effects (freshwater input to the ocean). These effects occur globally, but there is also significant regional variability in sea level change on this timescale due to vertical land motion.

NOTE: Short version of the same section, which maybe gets the point across more concisely.

Sea level is not a single measurement, but is rather a dynamic variable that

changes over a range of temporal and spatial scales (Woodworth et al., 2019). On short timescales (minutes to days), waves, tides, and storm surges dominate sea level variability (Pugh and Woodworth, 2014). On intermediate timescales (weeks to years), seasonal changes in ocean temperature and salinity drive sea level variability (Gill and Niller, 1973), as do changes in ocean circulation (Bingham and Hughes, 2009) and interannual climate variability (Han et al., 2017). On longer timescales (decades to millennia), sea level is influenced by changes in the Earth's climate and in the solid Earth (Fox-Kemper et al., 2021).

For the purposes of this thesis, it is convenient to separate sea level variability into two components: mean sea level, which we define as the long-timescale sea level trend, and the short and intermediate-term variability, which is the deviation from the mean state. When peaks in this variability coincide, they interact in a non-linear way to produce extreme sea level events (Idier et al., 2019). Long-timescale trends in mean sea level, such as those driven by anthropogenic climate change, change the severity of extreme sea level events or the frequency with which they exceed a fixed threshold (Boumis et al., 2023). Although this thesis concerns itself only with long-timescale sea level change, it is important to understand that the impacts of sea level rise are not felt in a binary way (i.e. land lies either above or below sea level), but rather as a continuously increasing risk to which coastal communities are exposed.

1.1.2 Drivers of long-term sea level change

1.1.3 Sea level and coastal communities

The IPCC's Sixth Assessment Report (AR6) highlights sea level rise as "an existential threat for island nations, low-lying coastal zones and the communities, infrastructure, and cultural heritage therein" (Cooley et al., 2022).

A rise of 1 m would increase the global population living below mean sea

level from 34–49 million people today to 77–132 million, depending on the choice of digital elevation model and population dataset (Seeger and Minderhoud, 2026).

1.2 The Antarctic Ice Sheet

The Antarctic ice sheet is the largest store of freshwater on Earth, holding **TODO: value** Gt of ice over an area of **TODO: value** km². It holds the potential to raise sea level by 57.25 m if it were to melt completely, although this is not considered likely in any future scenario of climate change (Fox-Kemper et al., 2021). Antarctica is situated over the South pole, bounded on all sides by the Southern Ocean. At its center, the ice sheet reaches altitudes of 4 km above sea level. This gives Antarctica one of the most extreme climates on Earth, with record minimum temperatures of -89.2°C (Turner et al., 2009). This section reviews the key regions of Antarctica, important changes to these regions in the recent past, and evidence of ice sheet change in the deep past (palaeoclimates).

1.2.1 Key regions of Antarctica

Antarctica is split into three climatically distinct regions: West Antarctica, East Antarctica, and the Antarctic Peninsula. The West Antarctic Ice Sheet (WAIS) is a 'marine' ice sheet, meaning that it is grounded on bedrock that lies predominantly below sea level. The deepest points of the bed are the Byrd subglacial basin and Bentley subglacial trench, reaching depths of 2.5 km below sea level. West Antarctic glaciers drain mainly into the Amundsen Sea, Ross Sea, and Weddell Sea sectors of the Southern Ocean. The latter two feature Antarctica's two largest floating ice shelves, the Ross ice shelf (**TODO: value** km²) and the

Filchner-Ronne ice shelf (**TODO: value** km²), respectively. The ocean cavities beneath them are key regions for the production of dense deep waters that connect Antarctica to the rest of the world through the global ocean circulation (Rintoul et al., 2025). **TODO: introduce Thwaites glacier here.**

The East Antarctic Ice Sheet (EAIS) currently stores the largest volume of ice in Antarctica and represents the majority of Antarctica's potential to raise sea level. It is separated from West Antarctica by the Trans-Antarctic mountains, the highest of which is Mt Kirkpatrick at 4,528 m above sea level. It is largely a terrestrial ice sheet (grounded above sea level), but it has three key marine basins: The Recovery, Wilkes, and Aurora subglacial basins. Each has a sea level potential on the same order of magnitude as the entire West Antarctic Ice Sheet **TODO: cite**. East Antarctica's Amery ice shelf, the third largest ice shelf in Antarctica (**TODO: value** km²), is the floating extension of the Lambert-Amery glacier system, Antarctica's largest drainage basin. East Antarctica is also home to Lake Vostok, the largest of the continent's subglacial lakes, with liquid freshwater that is up to 1 km deep (Siegert et al., 2001).

The Antarctic Peninsula Ice Sheet (APIS) is a comparatively small part of Antarctica, but is climatically distinct as the most Northerly region of the continent, reaching 68°S towards the Southern tip of South America. Its mountain range features smaller, Alpine-like glaciers down steep terrain, which contrast with the wide, low-gradient glaciers of the WAIS. The mountains act as a 2000 m orographic barrier between the Weddell Sea and the Bellinghausen Sea, blocking the prevailing Westerly winds and creating a steep East-West temperature and precipitation gradient across the peninsula (Schwerdtfeger, 1975; Van Lipzig et al., 2004). The largest ice shelf in the APIS, the Larsen ice shelf, extends down the Eastern flank of the peninsula and is divided into smaller sections Larsen A to D, the largest of which is Larsen C at **TODO: value and**

cite km².

1.2.2 Recent observed changes

Antarctica is not in equilibrium with the current climate, with some regions showing a clear trend in mass balance that is consistent across multiple methods of measurement (Gardner et al., 2018; Rignot et al., 2019; Schröder et al., 2019; Shepherd et al., 2019; Smith et al., 2020; Velicogna et al., 2020; Wang et al., 2021) **NOTE: I'm struggling to find any more recent papers**. The West Antarctic Ice Sheet is the main contributor to mass loss over this period, contributing 6.6 mm to global mean sea level between 1992–2020 (Otosaka et al., 2023). The vast majority of this mass loss has originated from the Amundsen Sea Sector of West Antarctica, particularly the Thwaites and Pine Island Glaciers. Sediment records from the Amundsen Sea indicate that Thwaites and Pine Island Glaciers have been in disequilibrium since at least the 1940s **TODO: cite**. Direct measurements of Pine Island Glacier made possible by the satellite era reveal a consistent pattern of change: retreat of the glacier terminus, acceleration of its ice stream, and a subsequent thinning as the glacier adjusts to maintain its flux balance (Joughin et al., 2003; Payne et al., 2004) **TODO: cite more recent**. It is difficult to draw direct links between the current disequilibrium and anthropogenic climate change, but it's likely that climate change has at least exacerbated the retreat of Pine Island glacier over this period (Bradley et al., 2025). **TODO: See Ines's 2023 remote sensing review paper for more inspiration on Pine Island. TODO: Thwaites**

Unlike the Arctic, which has warmed by **TODO: value** times the global average due to nonlinear feedbacks (Previdi et al., 2021), Antarctic surface air temperatures have broadly followed the global trend **TODO: cite**. The one exception to this is the Antarctic Peninsula, which has warmed three times faster

than the rest of the continent, up to temperatures otherwise unprecedented in the Holocene (Vaughan et al., 2003). This atmospheric warming has increased surface melting of the Antarctic Peninsula Ice Sheet and its resulting sea level contribution (Vaughan, 2006). The most pronounced changes in the APIS have been the thinning, retreat, and collapse of some of the peninsula's ice shelves (Cook and Vaughan, 2010; Rignot et al., 2013). The extent of ice shelves on the peninsula roughly trace the -9°C isotherm, suggesting a "limit of viability" for APIS ice shelves that will shrink with further warming (Morris and Vaughan, 2013). Although this is a useful heuristic, the mechanisms of ice shelf collapse are not purely thermal (see, for example, Sections 1.3.2 and 1.3.3).

The first ice shelf collapse to occur during the satellite era was the Wordie Ice Shelf in 1989 (Doake and Vaughan, 1991). This was followed by a sequential collapse from North to South down the Eastern flank of the peninsula, starting with the Larsen inlet — also in 1989 — then the Prince Gustav and Larsen A ice shelves in 1995 (Cook and Vaughan, 2010). Finally, the Larsen B ice shelf progressively thinned between 1992 and 2001 before its eventual collapse over a few days in February 2002 (Shepherd et al., 2003). The Larsen C ice shelf has so far avoided collapse, but has continued to thin at a rate of -3.8 m per decade over increasingly large areas of the ice shelf (Paolo et al., 2015). Between August 2015 and July 2017, a 200 km crack developed along the ice shelf, leading to the calving of a large iceberg representing approximately 10% of the total ice shelf area (Hogg and Gudmundsson, 2017). Due to the loss of buttressing force provided by ice shelves (see Section 1.3.2), the collapse of each ice shelf on the peninsula was consistently followed by an acceleration and thinning of its upstream glaciers (Rignot et al., 2004; Glasser et al., 2011; Dømggaard et al., 2025).

The collapse of the North-East peninsula ice shelves cannot be uniquely at-

tributed to anthropogenic climate change, since sediment records indicate that the Prince Gustav and Larsen A ice shelves have previously collapsed during the mid-Holocene (Pudsey and Evans, 2001; Brachfeld et al., 2003). This is corroborated by written accounts of the Prince Gustav ice shelf by 19th century polar explorers suggesting that it had been in retreat since at least 1843 (Cooper, 1997). However, similar records from ocean sediments previously beneath the Larsen B ice shelf indicate that it has been stable throughout the Holocene, and therefore suggest that its recent collapse contains an anthropogenic signal (Domack et al., 2005).

1.2.3 Antarctica in past climates

The collapse of Prince Gustav cannot be uniquely attributed to anthropogenic climate change, since .

NOTE: leaving this subsection for now because it's not clear to me how much detail to go into

1.3 Ice sheet dynamics

1.3.1 Ice-atmosphere interactions

1.3.2 Ice-ocean interactions

1.3.3 Calving and hydrofracture

1.3.4 Subglacial hydrology

1.3.5 Ice sheet instability

1.3.6 Solid Earth effects

1.4 Ice sheet modelling

1.4.1 Approximations in ice flow models

1.4.2 Basal sliding

1.4.3 Surface mass balance

1.4.4 Sub-shelf melting

1.4.5 Glacial isostatic adjustment

1.4.6 Projections of the Antarctic Ice Sheet

1.5 Uncertainty Quantification

1.5.1

Chapter 2

Methods

This is the methods chapter.

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